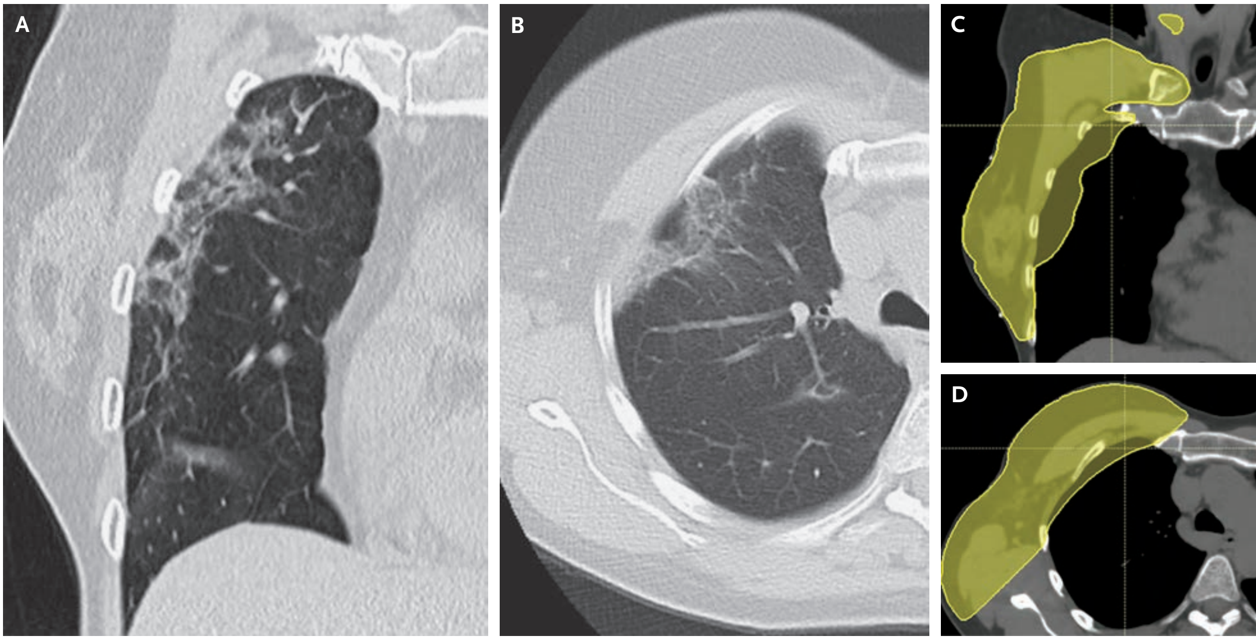
IMAGES IN CLINICAL MEDICINE

Radiation Pneumonitis



A 54-year-old woman with infiltrating ductal carcinoma of the right breast that had been treated with lumpectomy, locoregional radiation, and chemotherapy presented to the emergency department with a 4-week history of cough, fever, and dyspnea. Three weeks earlier, treatment with prednisone had been started owing to concern about radiation pneumonitis. However, the symptoms had flared with tapering of the glucocorticoid dose. On physical examination, an occasional cough was noted. Lung auscultation was normal. Computerized tomography (CT) of the chest showed fibrotic and ground-glass opacities in the irradiated area of the right upper lobe with volume loss and elevation of the minor fissure (Panel A, coronal view; Panel B, axial view) that corresponded to the radiotherapy field (yellow-shaded zone in Panel C [coronal view] and Panel D [axial view]). A diagnosis of radiation pneumonitis was made. The risk of radiation pneumonitis increases with the dose. In this case, the mean dose of radiation to the lung (8.1 Gy) was higher than the typical upper limit (7.5 Gy) owing to irradiation of the regional lymph nodes and a radiation boost to the tumor bed. Treatment with a prolonged tapering dose of glucocorticoids was given. At a 4-month follow-up visit, the symptoms had abated. Repeat imaging 6 months after the initial CT showed minimal subpleural postradiation changes.

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